

# FIT3143

APPLIED 01 PRESENTATION

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# **Task 1**

# **Network Topology**

- ① Compare and contrast different network topologies
- ② The best network topology selection
- ③ Reasons of the selection



# Comparison of Network Topology

| Topology         | Linear Array | Star         |
|------------------|--------------|--------------|
| Node Degree      | 1 or 2       | 1 or $N - 1$ |
| Network Diameter | $N - 1$      | 2            |
| Bisection Width  | 1            | 1            |
| Arc Connectivity | 1            | 1            |

| Topology        | Linear Array    | Star              |
|-----------------|-----------------|-------------------|
| Total Cost      | $N - 1$         | $N - 1$           |
| Geographic      | Moderate - High | Moderate          |
| Communication   | Multi Hop       | Centralised       |
| Fault Tolerance | Low             | Extremely Fragile |



# Comparison of Network Topology

| Topology         | Complete  | Binary Tree          |
|------------------|-----------|----------------------|
| Node Degree      | $N - 1$   | 1, 2 or 3            |
| Network Diameter | 1         | $2\log((N + 1) / 2)$ |
| Bisection Width  | $N^2 / 4$ | 1                    |
| Arc Connectivity | $N - 1$   | 1                    |

| Topology        | Complete       | Binary Tree    |
|-----------------|----------------|----------------|
| Total Cost      | $N(N - 1) / 2$ | $N - 1$        |
| Geographic      | Very Low       | Moderate       |
| Communication   | Direct         | Hierarchical   |
| Fault Tolerance | Very High      | Low - Moderate |



# Comparison of Network Topology

| Topology         | 2D Mesh           | 2D Torus      |
|------------------|-------------------|---------------|
| Node Degree      | 2, 3 or 4         | 4             |
| Network Diameter | $2(\sqrt{N} - 1)$ | $\sqrt{N}$    |
| Bisection Width  | $\sqrt{N}$        | $2(\sqrt{N})$ |
| Arc Connectivity | 4                 | 4             |

| Topology        | 2D Mesh           | 2D Torus  |
|-----------------|-------------------|-----------|
| Total Cost      | $2(N - \sqrt{N})$ | $2N$      |
| Geographic      | High              | Low       |
| Communication   | Multi Hop         | Multi Hop |
| Fault Tolerance | Moderate - High   | High      |



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# Network Topology Selection

The chosen network topology for the EV charging simulation is a **Hybrid Topology** combining a **2D Cartesian Mesh** for inter-node communication and a **Star Topology** for base station coordination.

## 2D Mesh Topology

- **Geographical Realism:** Matches the Cartesian grid and immediate-neighbour communication.
- **Hardware Efficiency:** Maximum 4 neighbours per node with multiple paths.

## Star Topology

- **Low-Latency Alerts:** Direct communication between EV nodes and the base station.
- **Delegating The Bottleneck:** High traffic is handled by the powerful base station.



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# Network Topology Selection

Why other topologies fail?

- **2D Torus:** Requires unrealistic wraparound connections.
- **Complete:** Requires excessive  $O(N^2)$  connections.
- **Linear Array:** High network diameter causes communication delays.
- **Binary Tree:** Link failures can disconnect entire subtrees.
- **Pure Star:** Creates a bottleneck and violates local-neighbour communication.



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# **Task 2**

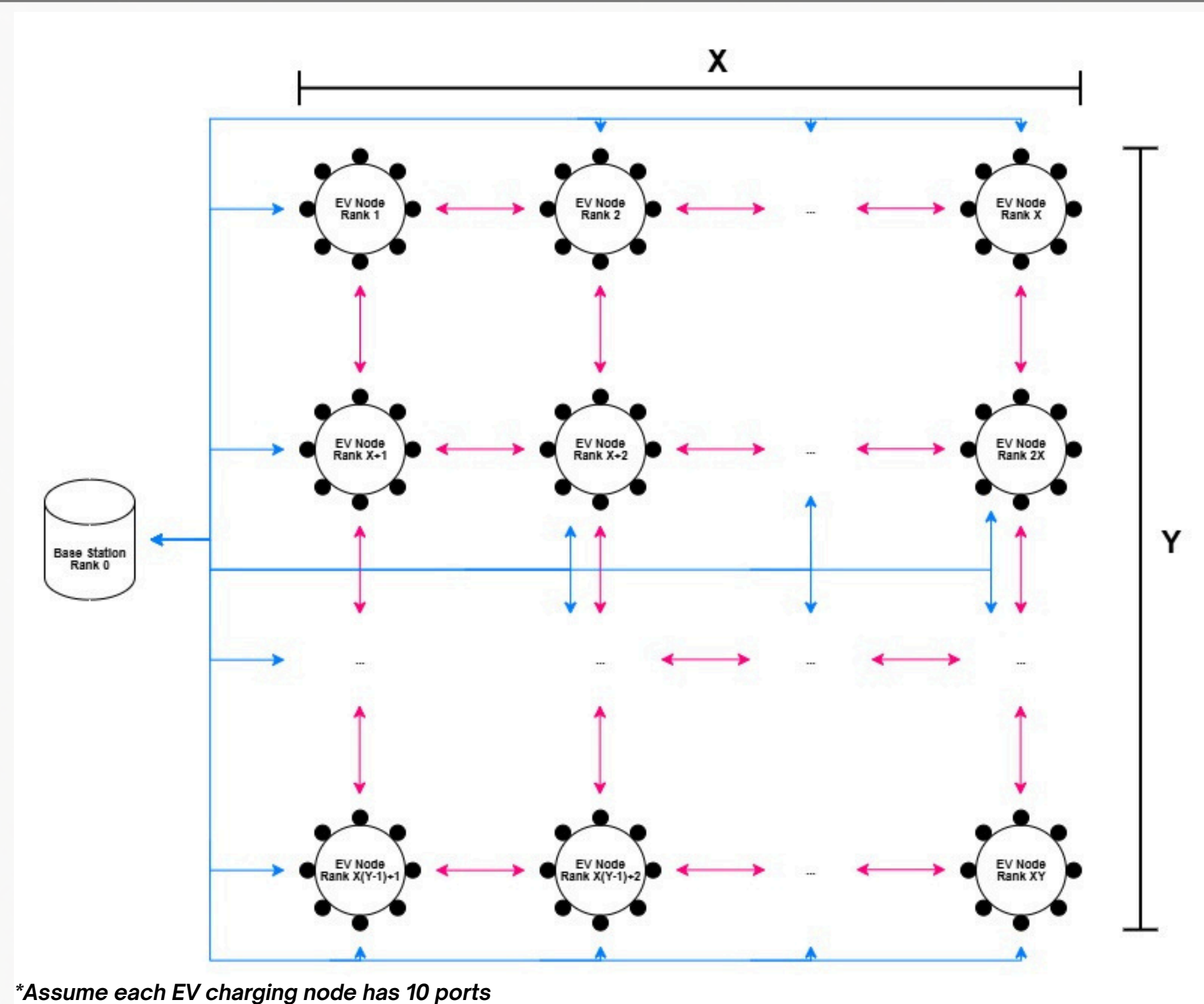
## **Architecture Design**

- ① Simulation of the Base Station, Charging Nodes, and Charging Ports
- ② Interactions among Base Station and Charging Nodes by Open MPI
- ③ Hardware Requirements
- ③ Network Bandwidth



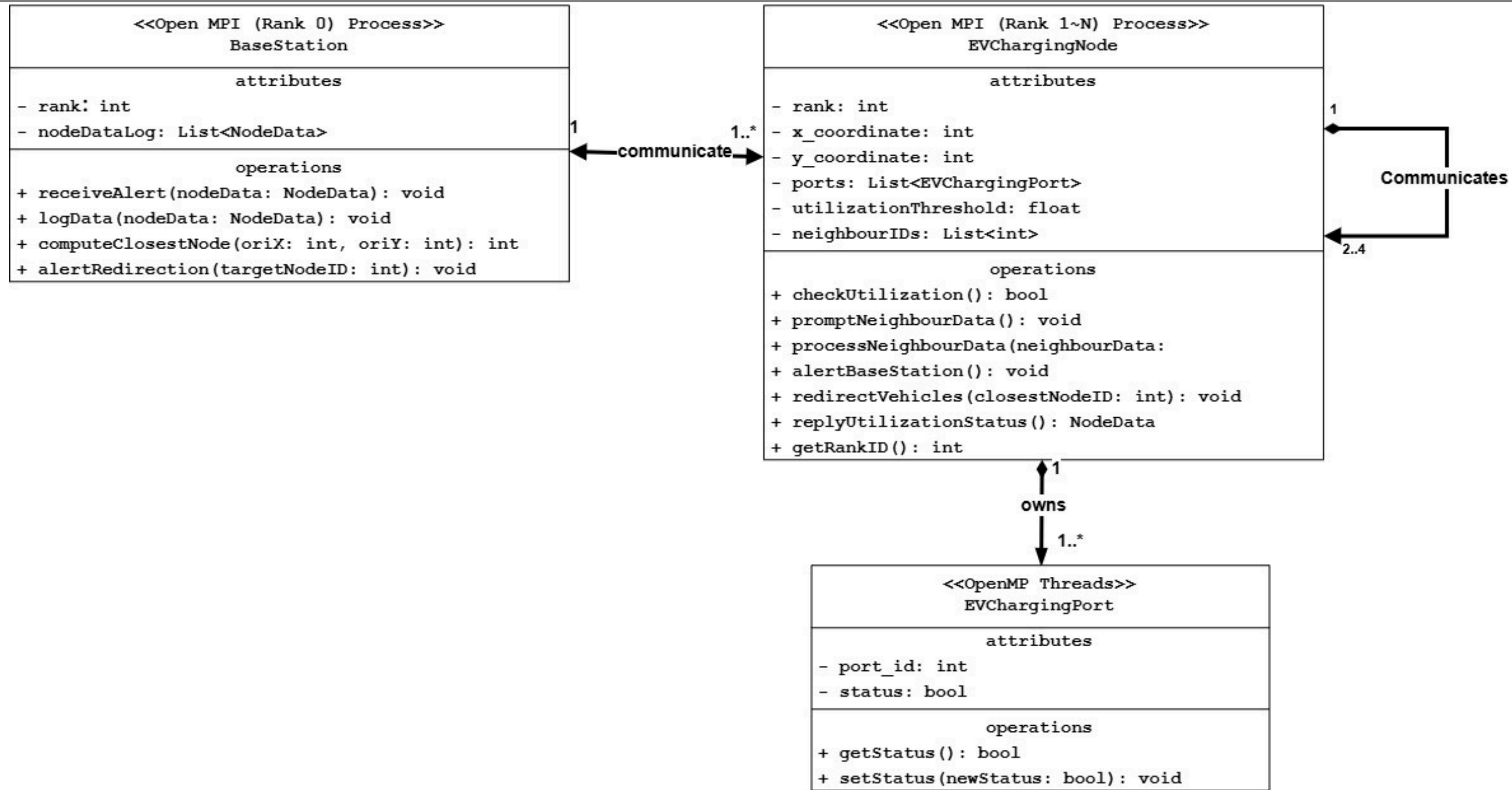
# Overview

- Combination of **2D Mesh** and **Star** topologies.
- **Red arrows** connect adjacent EV charging nodes for localized status probing, ensuring high scalability.
- **Blue arrows** connect all nodes to the Base Station exclusively for global redirection alerts.
- **Prevents** the Base Station from becoming a **communication bottleneck** by resolving minor congestions locally.





# Class Diagram





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# Class Diagram

Decoupling global routing, local processing, and physical ports.

## Base Station (Server)

- Operates as a single centralized at Rank 0
- Logs network data to monitor overall system health
- Computes the closest available node using spatial coordinates
- Sends redirection targets back to overloaded nodes

## EV Charging Node

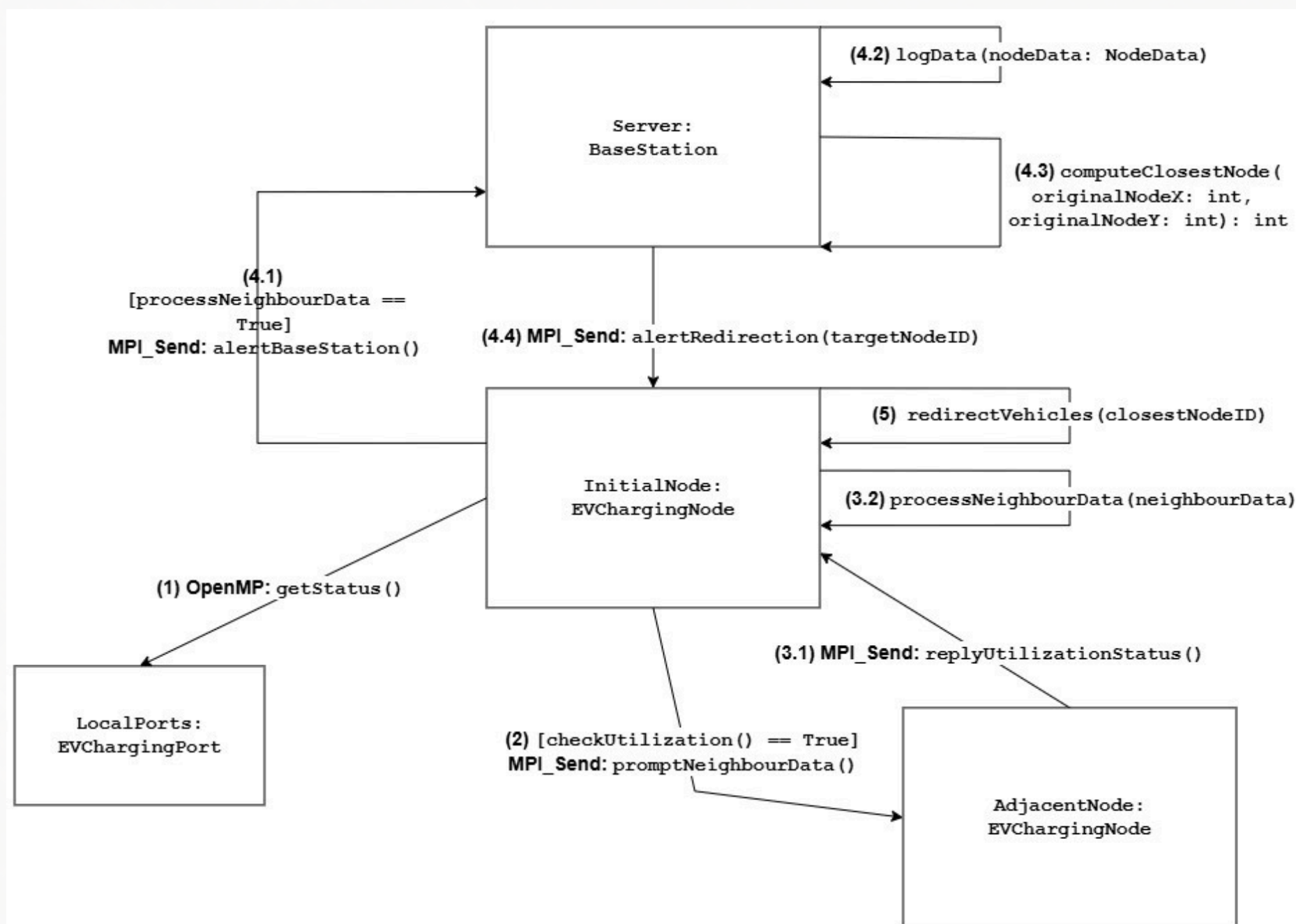
- Acts as an independent distributed MPI process
- Connects with adjacent orthogonal neighbors in the grid
- Caculates local utilization to determine congestion
- Probes neighbors to resolve minor traffic locally

## EV Charging Port

- Belongs strictly to one parent charging node
- Simulates the boolean occupancy status of a slot
- Uses OpenMP threads for concurrent status monitoring
- Eliminates network overhead via internal state checks



# Communication Diagram





# Communication Diagram

## Phase 1: Local Monitoring

- OpenMP threads call **getStatus()** to monitor charging ports concurrently
- Node evaluates local congestion by executing **checkUtilization()**

## Phase 3: Global Escalation

- Node escalates to the server using **MPI\_Send: alertBaseStation()** only when all neighbors are also overloaded

## Phase 2: Neighbor Probing

- High utilization (**threshold  $\geq 8/10$  ports**) triggers **MPI\_Send** via **promptNeighbourData()** to query adjacent nodes
- **promptNeighbourData()** aggregates replies to resolve minor congestion locally

## Phase 4: Centralized Redirection

- Base Station executes **computeClosestNode()** to find an available charging node
- Dispatches target ID via **MPI\_Send: alertRedirection()**, triggering **redirectVehicles()** at the node



# Assumptions

## Grid Layout

3 x 3 Cartesian grid  
9 EV Nodes + 1 Base  
Station

## Thread Mapping

1 MPI Process/Node  
10 OpenMP  
Threads/Node

## Topology

Nodes strictly  
communicate with 2-4  
immediate orthogonal  
neighbors

## Payload

Lightweight messages  
(64 bytes/message)

## Worst-case Frequency

1 alert event  
per second

## Safety Margin

10x of calculated  
application traffic



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# Hardware Requirements

Allocating dedicated CPU cores for every thread and process to avoid context-switching overhead to ensure maximum parallel efficiency.

## Hardware Allocation

- **OpenMP Requirements:** 9 Nodes x 10 Threads = 90 Cores
- **Base Station:** 1 Core
- **Total Compute Demand:** 91 Dedicated Cores

## Cluster Setup

- **Physical Machines:** 3 x 32-core machines = 96 Total Cores
- **Overhead Buffer:** Remaining 5-6 cores reserved for OS scheduling and system overhead



# Network Bandwidth

Calculating the maximum possible traffic during a simultaneous overload accurately determines the exact network bandwidth required.

## Worst-Case Traffic Calculation

- **Scenario:** All 9 nodes are overloaded simultaneously (1 event/sec)
- **Total Messages:** 66 messages/sec
  - 4 Corner nodes x 2 Neighbors = 8 Msgs
  - 4 Edge nodes x 3 Neighbors = 12 Msgs
  - 1 Center node x 4 Neighbors = 4 Msgs
  - Total Msgs = 24 Msgs x 2 (Request & Reply) = **48 Msgs**
  - 9 nodes ↔ base station = 9 Msgs + 9 Msgs = **18 Msgs**
- **Throughput:** 66 msgs x 64 bytes = 4224 bytes/s  $\approx$  34 kbps

## Bandwidth Recommendation

- With a 10x network safety margin, required bandwidth is  $\approx$  0.34 Mbps
- A minimal 1 Mbps Ethernet link is more than sufficient for this simulation



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# **Task 3**

## **Analysis**

- ① Transmission Delay Analysis
- ② Scalability & Complexity Evaluation
- ③ Architectural Optimization Strategy



# Assumptions

## Network Environment

- Simulated within a localized High-Performance Computing (HPC) cluster
- Physical distance is minimal; Propagation Delay is negligible

## Metric Isolation

- Analysis strictly isolates Physical Transmission Delay
- Software startup overhead, OS context switching, and CPU processing time are excluded the Network Interface Card bottleneck.

## Protocol Efficiency

- Assumes Open MPI over high-speed interconnects (e.g., InfiniBand with RDMA)
- Standard TCP/IP header bloat is bypassed; payload strictly maintained at hardware limits

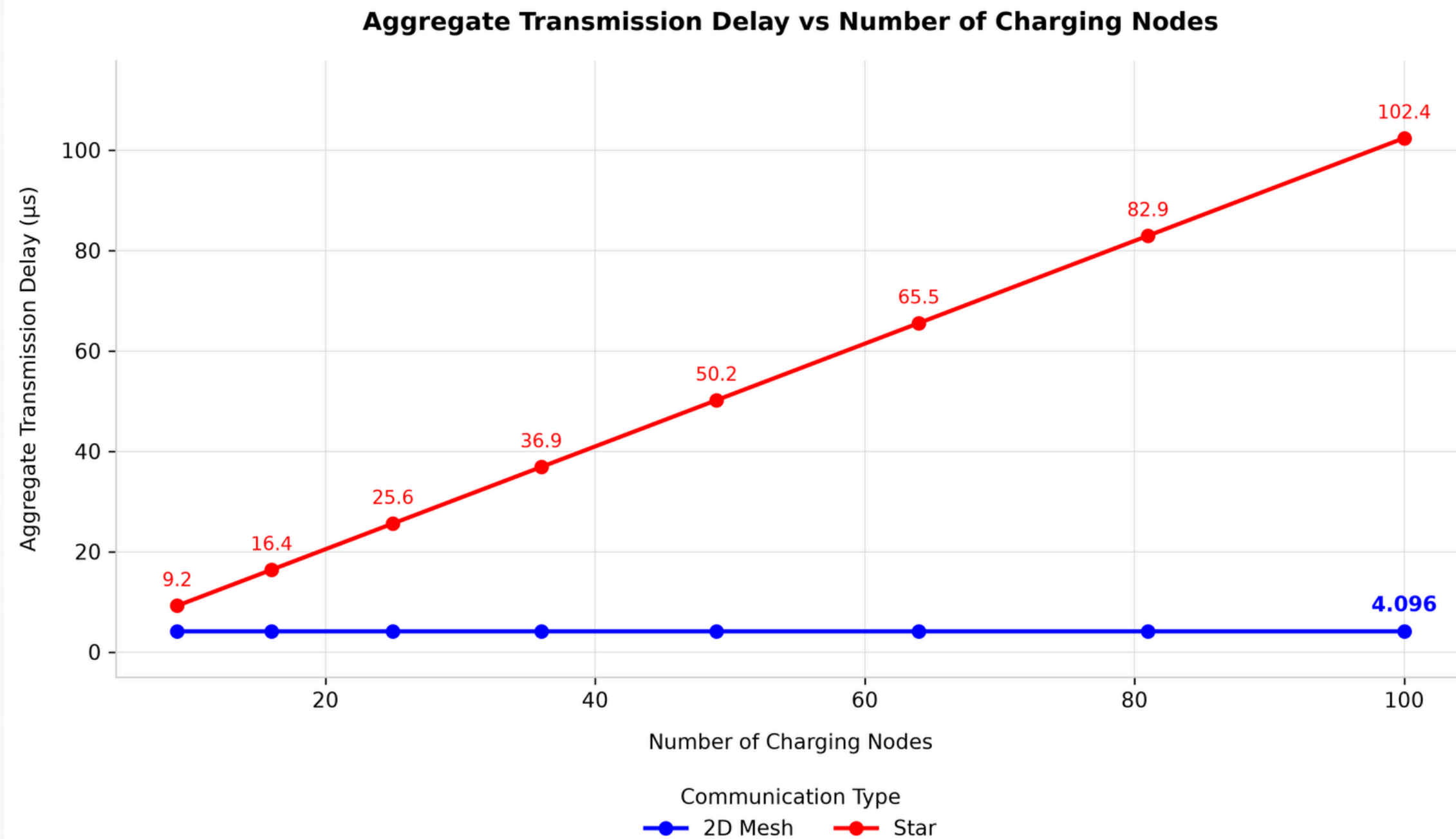
## System Constants

- Message Size (L): 512 bits (64 bytes)
- Network Bandwidth (B): 1 Gbps

- Base Transmissions Time (T):  $T = \frac{L}{B} = 0.512 \mu s$



# Transmission Delay Analysis





# Transmission Delay Analysis

Evaluating the theoretical time complexity and scalability bounds of the hybrid communication architecture.

## 2D Mesh Topology

- **Node Degree Distribution (N x N grid):**  
Corner Nodes (Degree 2): 4  
Edge Nodes (Degree 3):  $4N - 8$   
Inner Nodes (Degree 4):  $(N - 2)^2$
- Each local probe requires a strictly bounded **Request + Reply**
- Due to parallel execution, the total transmission delay is strictly bounded by the maximum node degree, establishing an  **$O(1)$  complexity** bound with a **constant maximum delay of  $4.096 \mu s$**

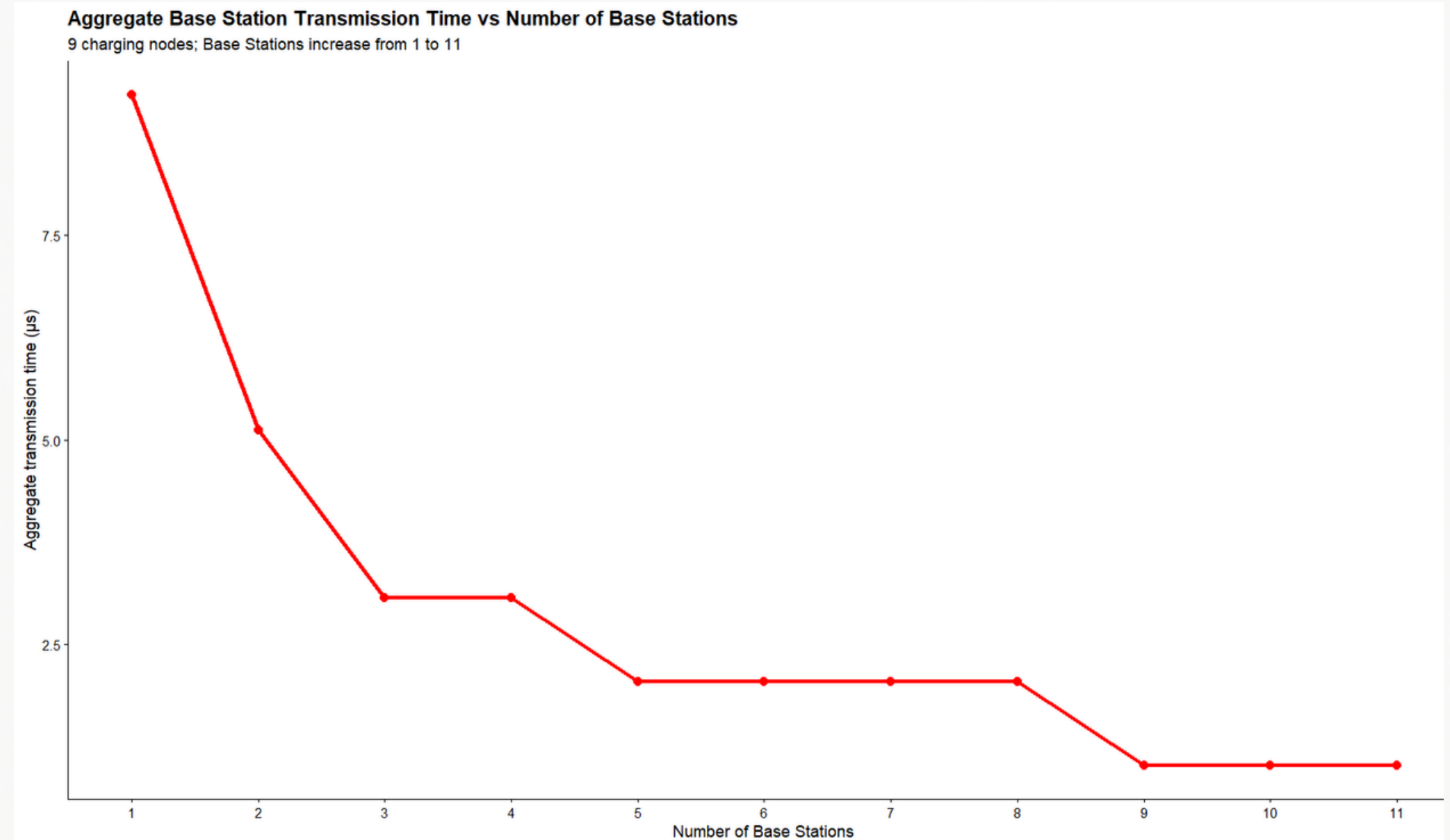
## Star Topology

- All N charging nodes funnel their round-trip alerts through a single Base Station's Network Interface Card
- This centralized design creates a fan-in bottleneck that forces concurrent alerts into **sequential queueing** at the server
- The aggregate delay scales linearly with the total number of nodes, yielding an  **$O(N)$  complexity** of  **$1.024N \mu s$**



# Transmission Delay Analysis

- Distributing nodes across  $S$  servers achieves an  **$O(N/S)$  load distribution**, effectively eliminating the centralized bottleneck
- The delay curve flattens toward an **asymptotic lower bound** constrained by the  $1.024\ \mu\text{s}$  physical hardware limit
- To avoid diminishing returns, configuring  **$N/4$  Base Stations** provides the optimal balance between latency and hardware cost

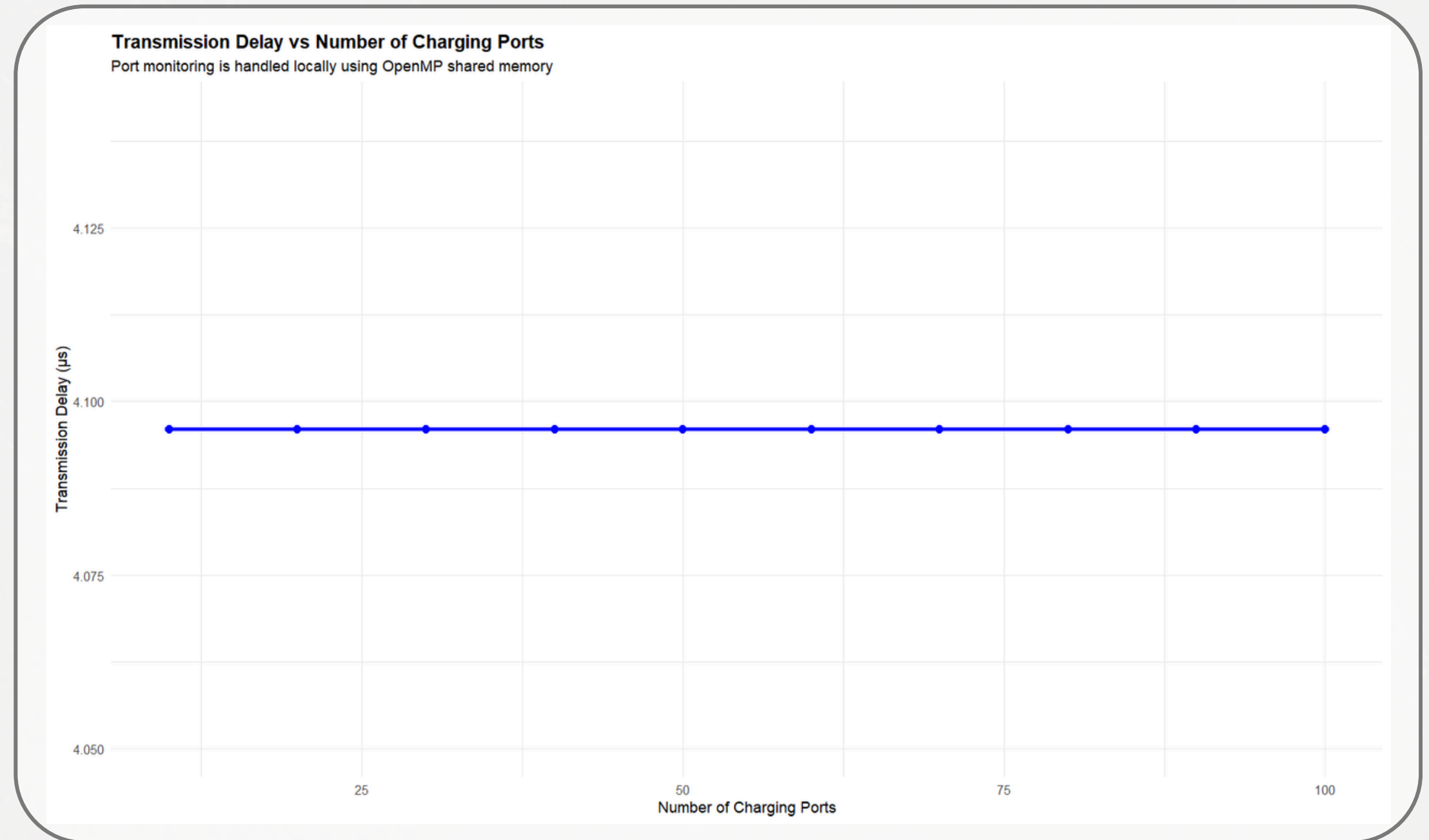


*\*Assuming the workload is balanced*



# Transmission Delay Analysis

- **Local Shared Memory:** Ports are managed via OpenMP threads, requiring zero MPI network communication
- **Zero Network Overhead:** Scaling the number of ports does not add any network transmission delay
- **O(1) Complexity:** Network transmission delay remains strictly constant at 0.512  $\mu\text{s}$





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# Summary.

- ① Conclusion
- ② Limitations
- ③ Future Works
- ③ Q & A



# Conclusion

- **Task 1:**

- Hybrid topology 2D with Star topology satisfies both wireless sensor network (WSN) geographic constraints and centralized logging needs
- Star overlay prevent multi hop latency degree  $\leq 4$  ensure realistic

- **Task 2:**

- Hybrid MPI + OpenMP enables distributed communication and local parallel processing.
- OpenMP threads handle charging ports concurrently without network overhead.
- MPI processes are mapped to dedicated cores to reduce contention.

- **Task 3:**

- More nodes  $\rightarrow$  higher BS workload; single BS is the bottleneck.
- More BS  $\rightarrow$  lower aggregate delay  $O(N/S)$
- 2-D mesh stays  $O(1)$  with max 4 neighbours.
- More ports have minimal network impact via local OpenMP.



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# Future Works

- Compare different alert frequencies to study how increased network traffic affects aggregate transmission time.
- Test different message sizes and bandwidths to evaluate their impact on transmission delay using L/B.



# The End

THANK YOU FOR LISTENING

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